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I declare that all material in this assignment/project/tutorial is my own work except where there is clear acknowledgement or reference to the work of others, and I have complied with and agreed to the University statement on Plagiarism and Academic Integrity.

Signature:

Date: 20/9/2026

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1. Case background

Starbucks Rewards is one of the world's largest retail loyalty programs and is currently undergoing a deliberate strategic shift. On 10 March 2026, Starbucks launched the program with 3 levels, Green, Gold & Reserve, and introduced a 60 Star redemption providing \$2 off any purchase in response to member requests for quicker access to rewards. Starbucks announced a record-high number of 35.6 million 90-day Active US Members, an increase of 4 % vs. prior years. The new \$2 reward was used by roughly one-third of all redemptions (Starbucks Corporation, 2026; Customer Experience Dive, 2026).

It has been explicitly stated in CEO Brian Niccol's view of the redesigned program that the primary focus should be on member engagement and personalization versus being "a coupon book" relying on discounts (Customer Experience Dive, 2026). The decision to redesign the program occurred after a documented reduction in promotional activity -- In October 2024, the Wall Street Journal reported that Starbucks had quietly decreased its promotions during the past year (Haddon, 2024).

Reducing the amount of promotional offers will only be sustainable if Starbucks can identify the specific customers that require incentives. A decrease in promotional offers without identifying customer requirements reduces the incentive for customers who are using the incentives provided while maintaining the subsidy to customers who have no intention of changing their purchasing behavior. Therefore, segmentation is the capability required for this stated strategy.

2. Business challenge

Starbucks cannot tell the difference with certainty in its Reward Members' buying behaviors when they are affected by a promotional offer or if their behaviors would have been the same without a promotion. Because these groups appear identical in the reporting data, promotional spending at the same time increases margins for non-incremental completions and understates how many members do see changes in their visit frequency (while cuts in promotional volumes increase the cost to target poorly).

This problem comes from measurement, not intent. The fact that a reward is marked successful simply because the customer has reached the required amount of spending

to get the rewards discount means that a member may be able to complete a program they never even opened. Thus, any performance metric created using "completions" as a criterion will count the first case as successful as well as the second. Likewise, any optimization algorithm optimized on "completion" metrics will continue to send promotions to members who were never influenced. That the completion of a reward is the incorrect objective function was the organizing premise behind our analyses.

3. Stakeholder and decision analysis

Leadership of loyalty programs is responsible for program economics and will decide which customer segments to support. Brand leadership is responsible for personalization as a promise to customers and wants evidence that it is a method (process) versus just a slogan. The CRM Team manages targeting and the choice of channels, and therefore needs segment labels that they can push into their campaign management systems. The finance group bears the cost of redemption and wants to know where the discount is coming from, and whether there is behavioral change. Retail store partners bear the burden of throughput of promotional activity. Members care about relevance and not being unfairly profiled, and the data governance of the lawful basis for automated targeting.

These six decisions define what the analysis needs to produce: What offer type does a member receive? Do members receive an offer at all? Which channel delivers the offer? How difficult is it to get the offer? Where is budget allocated across segments? And do some members need reengagement instead of a discount? If a segment could not make changes to one of these decisions, then that segment was considered to have no business value.

4. Proposed data and analytical method

4.1 Data

We used a publicly available dataset that simulates customer behaviour on the Starbucks Rewards mobile application: offer metadata for ten offers, demographic profiles for 17,000 members, and an event log of 306,534 records covering offers received, viewed and completed, and transactions. The dataset is simulated and is not a release of Starbucks' operational customer records. The company, the programme and the targeting problem are real and publicly documented; the transaction records

are simulated representations of that environment, so effect sizes below are illustrative of method rather than measurements of commercial performance.

Two data-quality issues shaped preparation. An age of 118 appears for 2,175 members and coincides exactly with null gender and income; it is a missing-data placeholder, and treating it as a real age would have shifted mean age from 54.4 to 62.6 years. These members were flagged rather than imputed. Separately, the event log stores its payload under different keys for different event types, so a naive flattening silently discards every completion record.

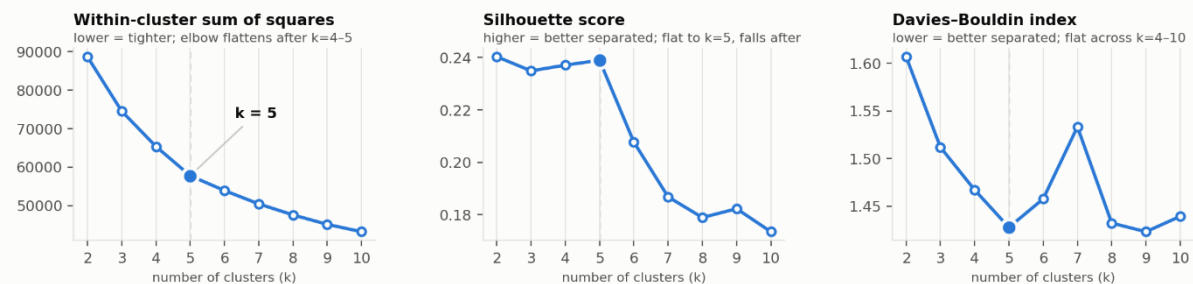
4.2 Feature construction

Because the distinction between influenced and uninfluenced completion was central to the question, we reformatted the offer funnel based on each offer. Every one of the 76,277 offers sent out then became an entry with its own valid date range. The logs of every single member were examined in chronological order to determine if there had been a view of this particular instance of the offer, whether it had been completed, and if the view had occurred prior to the completion. We determined what an influenced completion (the offer likely altered behavior) and what was an uninfluenced completion (reward delivered to a member who never looked at the offer). We have seven behavioral measures per member: view rate, influenced rate, uninfluenced rate, log spend, transactions made through the platform, share of spending outside any offer window, and tenure.

We purposefully did not include demographic attributes. If we built segments by age, sex, etc., we would be doing demographic targeting structurally instead of incidentally. These are reserved for segment profiling, so that we can continue to assess how successful our efforts were.

4.3 Method selection

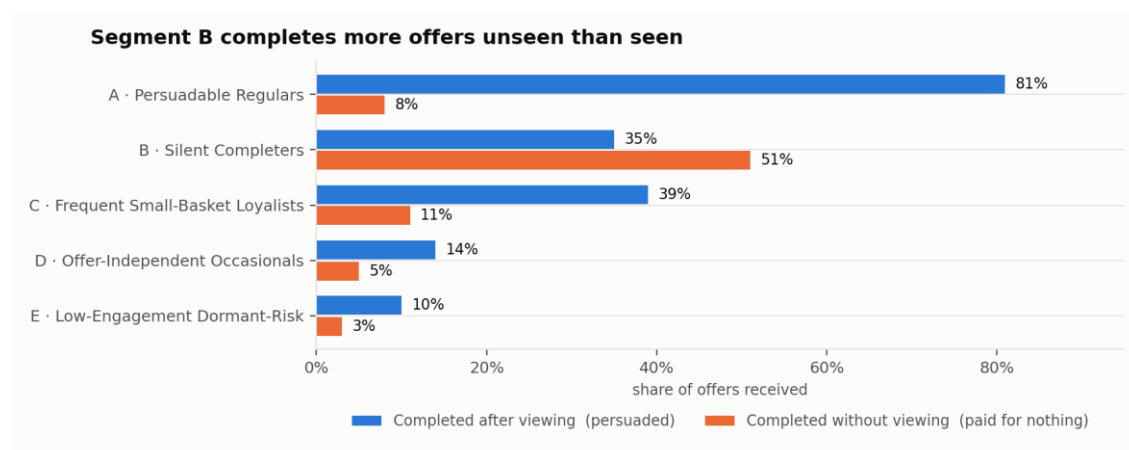
No k is statistically optimal: the criteria are flat between 4 and 5, so interpretability decides



Four approaches were evaluated on identical inputs rather than assuming the conventional choice. Rule-based RFM scoring was rejected as the primary method because the 30-day window makes recency uninformative and three quantile-cut dimensions cannot express the view-before-complete distinction. DBSCAN failed structurally: at a neighbourhood radius of 0.5, 89.4% of members were classified as noise, and at 1.3 and above, a single cluster absorbed all of them. Ward hierarchical clustering scored below K-Means at every k , with silhouette falling from 0.200 at $k=3$ to 0.157 at $k=7$.

K-Means was selected. No value of k was statistically optimal: silhouette at $k=5$ (0.239) lies within 0.001 of the $k=2$ maximum, and Davies–Bouldin varies only between 1.42 and 1.47 across $k=4$ to $k=10$. With the criteria indifferent between four and five clusters, interpretability decided — the five-cluster solution separates a group of frequent, small-basket members whose offer requirements differ materially from every other segment. Stability was strong: adjusted Rand index 0.998 across ten random initialisations and 0.990 across ten bootstrap resamples.

5. Expected results and business value



Of all 33,579 completed offers made by the membership, 23,487 (69.9 %) had been viewed prior to completion, and 10,092 (30.1%) were viewed after completion. In terms of total dollar value of rewards earned through the completion of offers, a total of \$164,676.00 was paid out for completing offers; of that amount, \$49,432.00, or 30.0% of the rewards paid out, went to members who never opened an offer.

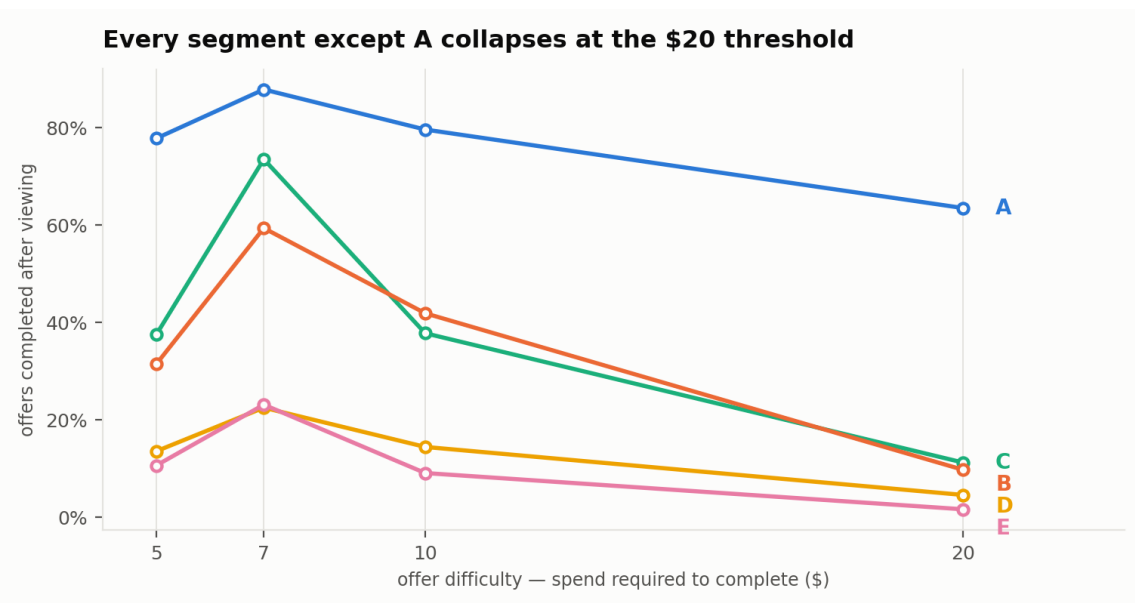
A. Persuadable Regulars (26.6%): The most active segment of the membership. They have viewed 96% of all offers sent to them and acted upon 81%. This segment is responsible for 44.5% of the revenue generated from the membership.

B. Silent Completers (21.5%): This group has spent almost as much as Persuadable Regulars but has completed 16% more offers without viewing than with viewing. They account for approximately \$32,267 in lost opportunities to increase sales.

C. Frequent Small-Basket Loyalists (14.3%): This group has the highest tenure with the program and is among the most frequent visitors. On average, they spend \$6.67 per visit.

D. Offer-Independent Occasionals (10.8%): The majority of this group's transactions occur during occasions when there are no special offers available to them. They generate 61% of their transactions outside of any offer windows.

E. Low-Engagement Dormant-Risk (26.7%): Members within this category tend to open offers; however, they rarely take action based on the information received in those emails. They have the lowest reward-to-revenue ratio compared to all other segments in the base.



Response varies sharply by offer design. Every segment except A collapses at the \$20 spend threshold; Segment C responds at 73.5% to a \$7 threshold and 11.2% at \$20. Channel produces the clearest single lever: social achieves the highest view rate in every segment, and for Segment B it raises the view rate from 56.9% on web to 84.8%, lifting the influenced rate from 0.310 to 0.441.

Segment → evidence → decision → measure			
Every segment earns its place by changing something Starbucks would otherwise do differently			
SEGMENT	EVIDENCE	DECISION	HOW IT IS MEASURED
A Persuadable Regulars 4,499 · 26.6%	View 96% · influenced 81% · \$175 spend · 44.5% of all revenue	Protect. Do not cut offer volume here during the discounting pull-back.	Influenced completion rate stays above 0.75
B Silent Completers 3,642 · 21.9%	51% of offers completed never viewed · holds \$32,267 of \$49,432 leakage · social lifts influence 0.31 → 0.44	Suppress spend-threshold discounts. Route remaining contact to social.	Reward released without a view, down from \$32,267
C Frequent Small-Basket Loyalists 2,426 · 14.3%	Basket \$6.67 · tenure 1,116 days · \$7 threshold 73.5% vs \$20 threshold 11.2% · discount 50.5% vs BOGO 28.4%	Low-threshold discount tier only. Retire \$20 offers and BOGO for this segment.	Influenced rate rising from 0.39 toward the 0.735 already seen at \$7
D Offer-Independent Occasionals 1,835 · 10.8%	61% of spend falls outside every offer window · lowest reward cost at 5.9% of revenue	Minimal intervention. Stop treating a self-sufficient segment as a promotion target.	Revenue held flat while offer volume falls
E Low-Engagement Dormant-Risk 4,526 · 26.7%	Views 71% but acts on 10% · 26.7% of members, 4.7% of revenue · worst reward-to-revenue at 11.9%	Change the threshold, not the volume: test \$5-\$7 entry offers on social before writing them off.	One cycle to lift influenced rate above 0.26, or reduce spend
ACROSS ALL SEGMENTS Replace "offer completion rate" with "influenced completion rate" as the programme KPI — 30.1% of today's completions are never seen by the member.			

Six are below. Remove B from spending threshold discounts; then send its other contacts to Social. Only give C low-threshold offers (the \$20 level and BOGO will need to go away) — this aligns with Starbucks' own findings, which show the company's \$2 redemption has become the program's most popular offer. Protect A from any across-the-board reduction in number of offers. Test an entry-level social promotion offer at a low threshold to E before reducing spend for one cycle. Take D out of promo targeting. Use Influenced Completion Rate (not Offer Completion Rate) as the measure the program reports. The first five recommendations address where rewards are currently allocated, while the sixth recommendation ensures the problem does not develop again.

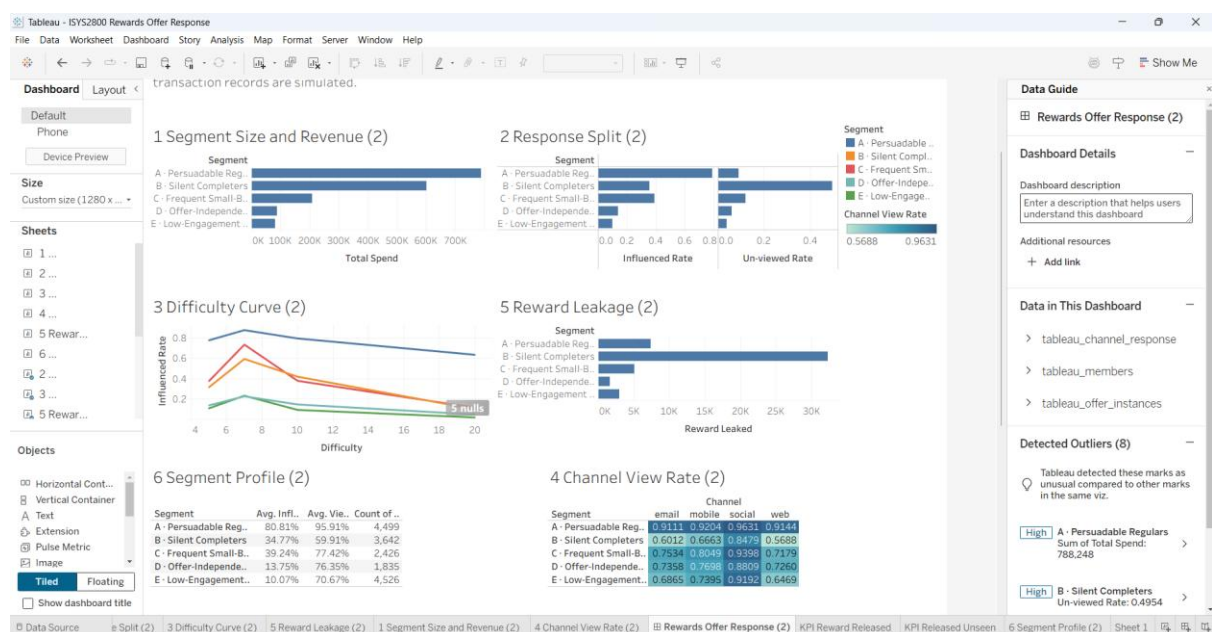
You need to pay close attention to the \$49,432. It represents reward dollars issued to customers without any indication of how those customers were persuaded or incentivized: these dollar amounts cannot be recovered because there is no data related to margin contained within the dataset, a portion of non-viewed completers may have been influenced by seeing the offer if they had viewed it; and since the data set did not contain a randomized control group — thus eliminating any possible causal

claims being made about the data what we can conclude is that the loss is significant, approximately 66% of that loss is concentrated into one identifiable segment of customer loyalty members, and the loss is quantifiable and therefore testable.

6. Digital solution

The prototype is an interactive Tableau desktop dashboard based on five extracts produced by the data analysis pipeline. There are four Key Performance Indicator (KPI) tiles with the data for member segments that have been identified, total value of rewards that were released to those member segments, number of times rewards were released, but no purchase was made (value released without a view), and how much of their sales are coming from rewards. There are six additional worksheets that display the data: Segment Size vs Revenue Contribution, Influenced vs Un-viewed Completion Rates, Response Rates vs Offer Difficulty, View Rate per Channel, Leakage By Segment, and a Segment Profile table.

All six worksheets are linked together through a single highlight action on the segment field. This means when you select a segment anywhere in the dashboard, it will be isolated throughout the rest of the dashboard. Therefore, if a loyalty manager wants to go from the overall leakage figure to the specific segment(s) causing this issue, they can do so in one click. The simulated-data disclosures are also located at the bottom of the dashboard and are therefore clearly visible and easily accessible for users viewing the dashboard.



7. Ethical, privacy and implementation considerations

Although the model did exclude demographics as an attribute of members, we can see how the resultant segments are income-graded — the average income range for Segment E was \$41,665 compared to Segment B with an average of \$72,271. Thus, behavioural segmentation creates a socio-economic gradient in terms of response rate — and thus offering discounts to high-response segments would target these benefits towards higher-income members. This is structural proxy discrimination — and excluding the attribute as a means of preventing it does nothing other than monitor the outcome to identify when it occurs. Therefore, we suggest auditing the income distribution of recipients of offers every cycle, although income is never an input into the modelling process.

A second finding supports this conclusion. The percentage of members with no demographic data ranged from 1.4 percent in Segment B to 27.1 percent in Segment E. Members that provide the least amount of demographic information have the highest likelihood of clustering within the lowest engaged segment. As such, any rules regarding demographic completeness will be inherently disadvantageous to these groups of members. Eligibility cannot rely solely upon demographics.

Regarding privacy, the segmentation utilizes only first-party data generated as part of the member's loyalty experience. This does not eliminate the issue of consent — members enroll in a loyalty program for the purpose of receiving rewards, not being denied access to them based on their own loyalty behavior. The way to mitigate against this is transparency. For example, if you withdraw a discount from a member who is already spending at a high frequency, providing a substitute reward or benefit instead of simply remaining silent would address this concern.

From a technical standpoint, implementing this is relatively straightforward. The inputs to the model are your organization's own event logs. The model fits in less than two seconds. The outputs of the model are a single label per member that could be easily written back to your CRM system. However, there are two primary risks associated with implementing this type of segmentation strategy. First, production data would require an identity resolution step that simulated data does not include. Duplicate and cross-device accounts would contaminate member-level features and lead to inaccurate results. Second, a segmentation strategy that relies on a single 30-day

window may drift over time. Particularly if there is some sort of major relaunch that alters redemption behaviors, then there will need to be periodic re-segmentation and ongoing drift monitoring to ensure continued accuracy. Accountability should fall to identified individuals in loyalty leadership for creating and managing the suppression list and the KPIs related to suppressing rewards, and analytics for identifying any drift in the segmentation strategy.

8. Conclusion

In order for Starbucks to deliver its stated intent to develop a Reward programme that will provide value through customer personalization as opposed to providing blanket discounts, the capability required to make this a reality is obscured by how Starbucks currently measures success. We found that nearly one-third of all completed offers were never viewed by the customer completing the offer. Behavioural segmentation of over 16,928 customers identified two-thirds of the unviewed activity within one segment; we also found a second group of loyal customers with very poor retention due to their high average spend. As such, our research provides six potential actions that Starbucks could take with regard to its Rewards Programme. The most impactful of those options is not an action based on how Starbucks targets individual customers (i.e., "which type" of customer should receive which offer), but instead a new metric for measuring performance - only count as complete each time a reward was successfully offered and viewed by the customer making the purchase. If Starbucks does not begin to use this metric, then the improved results from optimizing the rewards program will be measured against an incorrect baseline.

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